

What is Data?

Data is a collection of facts, observations, measurements, or raw information. Patterns and Observations can be learned from data, to draw meaningful relations and gain useful knowledge.

Data and Information are two different terms, while data is raw and unprocessed, information is the useful knowledge you obtain after processing the data.

Data Point is a single observation or example in a collection of data. Suppose for an employee list data, a datapoint would be 'name: Varun', 'position: Jr. Software Engineer', 'portfolio: Shabeeb'. These smallest individual units are called data points.

Records are collections of data points, used to give knowledge on a particular entity.

Variable is a property that can have different values, or are values which change throughout the lifecycle of the data.

Metadata is data which describes other data, for example, if we check properties of a file in our file system, we can see the file type, the space it occupies, whether it's read only or not, etc. These properties are metadata, as the actual data is within the file.

Data is the fundamental unit of AI systems, as models are trained upon large quantities of data, thus making it the fundamental building block of AI and its related concepts.